

DSA0405 - Fundamentals of Data Science

Total points 20/25

CO2-AT3 Assessment Tool MCQs

✓ 1. A dataset follows a **normal distribution** with $\mu = 500$ and $\sigma = 50$. What is 1/1 the probability that a randomly chosen value is **above 600**?

- ☒ a) 2.28%
- ☐ b) 5%
- ☐ c) 16%
- ☐ d) 0.13%



✓ 2. A dataset contains salaries that follow a **normal distribution** with $\mu =$ 1/1 **75,000** and $\sigma = 10,000$.

What is the **Z-score** for a salary of **95,000**?

- ☐ a) 1.5
- ☒ b) 2.0
- ☐ c) 2.5
- ☐ d) 3.0



- ✓ 3. A Hadoop cluster stores **500 GB** of data with a **block size of 128 MB** and **1/1** a **replication factor of 3**.

How much **total storage** is required in the cluster?

- ☐ a) 500 GB
- ☐ b) 1000 GB
- ☒ c) 1500 GB
- ☐ d) 2000 GB



- ✓ 4. A dataset contains **1 million records**, and **20% of them** belong to **Class A**. **1/1**
If we randomly **sample 5000 records**, what is the expected number of Class A records in the sample?

- ☐ a) 1000
- ☒ b) 1250
- ☐ c) 1500
- ☐ d) 2000



- ✓ 5. A MapReduce job processes **1 TB** of data using **100 nodes**. If each node **1/1** processes **10 GB per hour**, how long will the entire job take?

- ☐ a) 1 hour
- ☒ b) 2 hours
- ☐ c) 10 hours
- ☐ d) 20 hours



- ✓ 6. A dataset has **1000 samples**, where:
600 belong to Class A 400 belong to Class B

1/1

A split results in two child nodes:

Left: **300 Class A, 50 Class B** Right: **300 Class A, 350 Class B**

What is the **information gain**?

- ☐ a) 0.125
- ☐ b) 0.278
- ☒ c) 0.541
- ☐ d) 0.672



- ✓ 7. A dataset has **10 features**. PCA retains **95% variance** with the top **4 components**. What percentage of variance do the remaining components contribute?

1/1

- ☐ a) 2%
- ☒ b) 5%
- ☐ c) 10%
- ☐ d) 20%



✓ 8. A dataset contains values ranging from **1 to 1,000,000**. Which transformed value does **10,000** map to if we apply a **log transformation using base 10**? 1/1

- ☐ a) 2
- ☐ b) 3
- ☐ c) 5
- ☒ d) 4



✓ 9. A dataset follows a **normal distribution** with $\mu = 150$ and $\sigma = 20$. What percentage of values lie **beyond 3 standard deviations** from the mean (both tails combined)? 1/1

- ☒ a) 0.3%
- ☐ b) 1.35%
- ☐ c) 2.5%
- ☐ d) 5%



✓ 10. A dataset has **1 million records** and **5% of them** contain at least **one missing value**. If we remove **all rows** with missing values, how many records remain? 1/1

- ☐ a) 900,000
- ☐ b) 950,000
- ☐ c) 990,000
- ☒ d) 975,000



✓ 11. A classification model gives **Precision = 0.80** and **Recall = 0.60**. What is the **F1-score**? 1/1

- ☐ a) 0.65
- ☒ b) 0.70
- ☐ c) 0.75
- ☐ d) 0.80



✓ 12. Consider the following NumPy matrix:

1/1

$A = \begin{bmatrix} 2 & 4 & 3 & 6 \end{bmatrix}$

- ☒ a) 0
- ☐ b) 2
- ☐ c) 6
- ☐ d) 12



✓ 13. A **Pandas DataFrame** with **1 million rows** is sorted using `df.sort_values(by='column1')`. The sorting algorithm used by Pandas is:

1/1

- ☐ a) Merge Sort
- ☐ b) Quick Sort
- ☒ c) Tim Sort
- ☐ d) Heap Sort



✓ 14. A Matplotlib plot renders **1 million data points** using `plt.scatter()`. The plot is slow. What is the best way to optimize performance? 1/1

- ☐ a) Use `plt.plot()` instead of `plt.scatter()`
- ☐ b) Enable `marker="."` in `plt.scatter()`
- ☒ c) Use `ax.hexbin()` instead of `plt.scatter()`
- ☐ d) Increase figure size



✓ 15. A **Pandas DataFrame** has **10 million rows** and **100 columns**. Which method is **fastest** for selecting **specific rows** based on conditions? 1/1

- ☐ a) `df[df['column1'] > 100]`
- ☐ b) `df.loc[df['column1'] > 100]`
- ☒ c) `df.query('column1 > 100')`
- ☐ d) `df.apply(lambda x: x['column1'] > 100, axis=1)`



✗ 16. A **NumPy array** of shape **(100,000, 10)** is stored using **float64** data type. What is the total memory consumption in MB? 0/1

- ☐ a) 8 MB
- ☒ b) 80 MB
- ☐ c) 160 MB
- ☐ d) 320 MB



Correct answer

- ☒ c) 160 MB



✗ 17. A **K-Means clustering model** is run on **1 million points** with **K=5**. The convergence time is **10 minutes**. If **K is doubled to 10**, assuming all other factors remain the same, what is the expected increase in execution time? 0/1

- ☐ a) No change
- ☐ b) 2x increase
- ☒ c) 4x increase
- ☐ d) 10x increase

✗

Correct answer

- ☒ b) 2x increase

✓ 18. A **NumPy** array of shape **(200,000, 5)** is stored using **float32** instead of **float64**. How much memory is saved in MB? 1/1

- ☐ a) 2 MB
- ☐ b) 4 MB
- ☒ c) 8 MB
- ☐ d) 16 MB

✓



✗ 19. A dataset contains **100,000 data points**, and a **scatter plot** is generated 0/1 using **Matplotlib**. Each marker takes **100 bytes** of memory. What is the total memory usage for the scatter plot?

- ☒ a) 1 MB
- ☐ b) 5 MB
- ☐ c) 10 MB
- ☐ d) 100 MB

✗

Correct answer

- ☒ c) 10 MB

✗ 20. A **Pandas DataFrame** with **1 million rows** is sorted based on a **single column**. The sorting algorithm used by Pandas is **Timsort ($O(n \log n)$)**. If sorting takes **5 seconds**, how long would it take for **10 million rows**? 0/1

- ☐ a) 10 seconds
- ☐ b) 50 seconds
- ☐ c) 25 seconds
- ☒ d) 60 seconds

✗

Correct answer

- ☒ c) 25 seconds



✓ 21. A Pandas DataFrame contains **20,000 rows**. After applying a filter that removes **40% of the data**, how many rows remain? 1/1

- ☐ A) 10,000
- ☒ B) 12,000
- ☐ C) 14,000
- ☐ D) 16,000



✓ 22. A NumPy array contains **numbers from 1 to 500**. What is the mean of this dataset? 1/1

- ☐ • A) 245
- ☒ • B) 250
- ☐ • C) 255
- ☐ • D) 275



✓ 23. A dataset contains **5000 values** distributed into **50 bins** in a histogram. What is the average number of values per bin? 1/1

- ☐ A) 80
- ☐ B) 90
- ☒ C) 100
- ☐ D) 110



✓ 24. Which sub-package of SciPy is specifically designed for handling vector quantization and K-Means clustering algorithms?

- ☐ A) scipy.optimize
- ☐ B) scipy.spatial
- ☒ C) scipy.cluster
- ☐ D) scipy.linalg



✗ 25. What core data science task are both NumPy and SciPy explicitly built to solve together? 0/1

- ☐ A) Web scraping and API deployment.
- ☒ B) Mathematical and numerical analysis.
- ☐ C) Static data visualization and dashboarding.
- ☐ D) Database management and query optimization.



Correct answer

- ☒ D) Database management and query optimization.

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